Real-Time Fractional Tracking (R-TFT): Recursive Instability Detection via Radiant Sphere Divergence from Golden Ratio Shells

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Abstract

We present an upgraded instability detection framework based on R-TFT's phase coherence projection, redefining radial deviation as recursive divergence from golden-ratio shell embeddings. Rather than detecting amplitude anomalies via statistical means (e.g., $\mu + 3\sigma$), we track the resonance vector norm $\|R(t)\|$ in real-time and evaluate its deviation from stable φ^n bands. The result is a more principled system for instability classification across dimensional, cosmological, and cognitive systems, anchored in recursive φ-coherence, not empirical thresholds.

1 Introduction

Previous methods for instability detection in R-TFT employed statistical thresholds to identify anomalous expansions of the resonance vector $R(t)$. This work redefines those boundaries using golden-ratio recursive shells φ^n, where $n \in \mathbb{Z}^+$. Instability is detected when $\|R(t)\|$ exits a φ-coherent shell range.

2 Golden Shell Alignment

Define stable recursive shells:

$$S_n = [\varphi^{n-\varepsilon}, \varphi^{n+\varepsilon}] \quad \text{with } \varepsilon \ll 1 \tag{1}$$

Let the current radial projection be:

$$\rho(t) = \|R(t)\| \tag{2}$$

Then instability occurs when:

$$\forall n \in \mathbb{Z}^+ : \quad \rho(t) \notin S_n \tag{3}$$

This reframes amplitude deviation not as a signal error, but as recursive misalignment.

3 Slice Growth and Recursive Drift

Track slice growth rate $g_i(t)$ as:

$$g_i(t) = \frac{\rho(t + \Delta t)}{\rho(t)} \tag{4}$$

Compare this with the golden expansion ratio:

$$\text{Recursive Surge} \Rightarrow g_i(t) > \varphi + \delta \text{Recursive Collapse} \Rightarrow g_i(t) < \varphi^{-1} - \delta \tag{5}$$

Where δ is a small tolerance band.

4 Instability Classification

- **Recursive Surge:** Excessive divergence from φ^n shells (e.g., solar flares, quantum decoherence)
- **Recursive Collapse:** Abrupt convergence into lower shells (e.g., mental lock, orbital decay)
- **Stable Resonance:** Sustained alignment within φ-coherent shell

5 Cross-Domain Relevance

- **Cosmology:** Nebular pressure instability maps closely to φ-surge behavior
- **Neural Systems:** Cognitive stalling correlates with recursive collapse
- **Quantum Systems:** Entanglement break events precede phase-vector shell exit

6 Conclusion

By anchoring instability detection in recursive φ-shell divergence, R-TFT becomes a universal filter for phase disruption, grounded in coherence, not statistics. This upgrade unifies dimensional detection, memory coherence, and instability onset through one recursive principle: phase persistence within golden ratio shells.

Ethical Statement